Equation and Data Sheet PHYC10009 (may be detached)

Mechanics

$x(t) = v_\mathrm{i}t + \frac{1}{2}at^2$	$v(t) = v_\mathrm{i} + at$	$\Delta x = \frac{1}{2}(v_\mathrm{i} + v_\mathrm{f})\Delta t$
$\bar{v} = \frac{v + v_0}{2}$	$\bar{v} = \frac{\Delta x}{\Delta t}$	$v_\mathrm{radial} = \frac{2\,\pi\,r}{T}$
	$\bar{a} = \frac{\Delta v}{\Delta t}$	$2a\Delta x = v_\mathrm{f}^2 - v_\mathrm{i}^2$
$\vec{F} = m\vec{a}$	$F_\mathrm{fr} = \mu_\mathrm{k}F_\mathrm{N}$	$F_\mathrm{radial} = \frac{mv^2}{r}$
$F_\mathrm{g} = mg$	$F_\mathrm{G} = G\frac{m_1 m_2}{r^2}$	$\sum F = ma$
$F_\mathrm{S} = -kx$	$\sum \vec{p}_{before} = \sum \vec{p}_{after}$	$\vec{p} = m\vec{v}$
$\omega = \sqrt{\frac{k}{m}}$	$T = \frac{1}{f}$	
$W = \vec{F}\cdot\vec{d} = Fd\cos\theta$	$\tau = rF\sin\phi$	$\vec{J} = \int_{t_1}^{t_2}\vec{F}\,dt = \vec{F}_\mathrm{av}\Delta t = \Delta p$
$l_\mathrm{arc\ length} = r\theta_\mathrm{radians}$		$\frac{\theta°}{360°} = \frac{l}{2\pi r}$
$PE_\mathrm{g} = mgh$	$KE = \frac{1}{2}mv^2$	$PE_\mathrm{el} = \frac{1}{2}kx^2$
Simple harmonic motion	$x(t) = A\cos(\omega t) = A\cos\left(\frac{2\pi}{T}t\right)$ $v(t) = -A\omega\sin(\omega t)$ $a(t) = -A\omega^2\cos(\omega t) = -\omega^2 x(t)$	

Sound, Optics

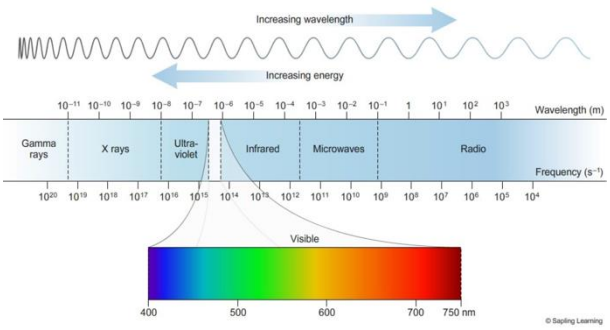
$v = \frac{c}{n}$	$v = f * \lambda$	$\beta_\mathrm{dB} = 10\log_{10}\left(\frac{I}{I_0}\right)$
$\frac{1}{f} = \frac{1}{d_0} + \frac{1}{d_\mathrm{i}}$	$n_1\sin\theta_1 = n_2\sin\theta_2$	$I = \frac{P}{A}$
$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$	$M = -\frac{d_i}{d_0}$ $\quad P = \frac{1}{f}$	$\frac{h_i}{h_0} = \frac{d_i}{d_o}$

Electricity, Electromagnetism

$E = k * \frac{Q}{r^2}$	$F = k\frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r^2}$	$V = k\frac{Q}{r}$
$PE = \frac{kqQ}{r}$	$PE = -qEd = q\Delta V$	$W = Fd = Eqd$
$I = \frac{Q}{t}$	$V = IR$	$R = \rho\frac{l}{A}$
$F = IlB\sin\theta$	$F = qvB\sin\theta$	$P = \frac{QV}{t} = IV$
$E = \frac{Q}{\epsilon_0 A}$	$\varepsilon = -N\frac{d\Phi}{dt}$	$\vec{\mu} = I * \vec{A}$
$\Phi = \vec{B} * \vec{A} = BA\cos\theta$	$V_S = N_S\frac{\Delta\Phi_B}{\Delta t}$ $V_P = N_P\frac{\Delta\Phi_B}{\Delta t}$	$R_{eq,series} = \sum R$
$V_{loop} = \sum V_i = 0$	$\sum I_{in} = \sum I_{out}$	$R_{eq,parall} = \sum\left(\frac{1}{R}\right)^{-1}$
$B_{loop} = \frac{\mu_0 N_p I}{2r}$	$B_{wire} = \frac{\mu_0 I}{2\pi r}$	$B_{solenoid} = \frac{\mu_0 I\, N}{l}$
$Q = C * V$	$C = \varepsilon_0\kappa * \frac{A}{d}$	$PE = \frac{1}{2} * C * V^2$

Atomic Physics, Radiation

$\frac{1}{\lambda} = R(\frac{1}{n'^2} - \frac{1}{n^2})$	$E_n - E_m = 13.6\,eV\left(\frac{1}{m^2} - \frac{1}{n^2}\right)$	$E_n = -13.6\,eV\,\frac{Z^2}{n^2}$
$PE = -eV = k\,\frac{Ze^2}{r}$	$\lambda = \frac{\ln(2)}{t_{1/2}}$ $\lambda_x = \frac{\ln(2)}{l_{1/2}}$	$N(t) = N_0 e^{-\lambda t}$
$E = mc^2$	$\Delta x * \Delta p \geq \frac{h}{2\pi}$	$R = N * \lambda$
$E = hf = \frac{hc}{\lambda}$	$\lambda = \frac{h}{p} = \frac{h}{mv}$	$R(t) = R_0\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$
$D_{abs} = \frac{\Delta E}{m}$	RBE alphas = 10-20 RBE beta = 1-2 RBE gamma = 1	$I(x) = I_0 e^{\left(-\frac{x}{\lambda_x}\right)}$



Electromagnetic Spectrum

Constants

Newtonian constant of gravitation	G	6.67×10^{-11} Nm^2/kg^2
Gravit. Accel. at Earth's surface	g	9.81 N/kg
Speed of light in vacuum	c	$3.00\ 10^8$ m/s
Speed of sound in air (20 °C, 1atm)	v	343 m/s
Permittivity of vacuum	ϵ_0	8.854×10^{-12} F/m (C^2/ Nm^2)
Electrostatic constant	k	8.988×10^9 m/F (N m^2/C^2)
Permeability of vacuum	μ_0	$4 * \pi * 10^{-7}$ Tm/A
Avogadro constant	N_A	6.022×10^{23} 1/mol
Atomic mass unit	u	1.661×10^{-27} kg 931.5 MeV/c^2
Electron Volt	eV	1.6×10^{-19} J
Charge of 1 e	e	-1.6×10^{-19} C
Planck constant	h	6.63×10^{-34} m^2 kg/s 4.136×10^{-15} eV s
Rydberg constant	R	1.0974×10^7 1/m
Radius of Earth	R_E	6.38×10^6 m
Mass of Earth	M_E	5.98×10^{24} kg
Radius of Sun	R_S	6.96×10^8 m
Mass of Sun	M_S	1.99×10^{30} kg
Distance Earth-Sun	D_{ES}	149.6×10^9 m
Distance Earth-Moon	D_{EM}	384×10^6 m
Proton mass	m_p	1.673×10^{-27} kg 1.007276 amu
Neutron mass	m_n	1.675×10^{-27} kg 1.008665 amu
Electron mass	m_e	9.109×10^{-31} kg 0.000549 amu
10^{-12}	p	pico
10^{-9}	n	nano
10^{-6}	μ	micro
10^{-3}	m	milli
10^0	–	–
10^3	k	kilo
10^6	M	Mega
10^9	G	Giga